01/28/2026

Homework - 1

Q1) Prove that there exists an integer n such that:

$$f(n) > f(n+1),$$

where $f(n) = n^3 - 303n^2 + 30600n - 1030000$

This is an existential claim, so we use proof by example method

Lets consider

$$f(n+1) - f(n)$$

$$\therefore f(n) = n^3 - 303n^2 + 30600n - 1030000$$

$$\therefore f(n+1) = (n+1)^3 - 303(n+1)^2 + 30600(n+1) - 1030000$$

$$= n^3 + 3n^2 + 3n + 1 - 303(n^2 + 2n + 1) + 30600n + 30600 - 1030000$$

Now, subtracting $f(n)$:

$$f(n+1) - f(n)$$

$$\therefore \cancel{n^3} + 3n^2 + 3n + 1 - \cancel{303n^2} - 606n - 303 + \cancel{30600n} - 999,400$$

$$- (\cancel{n^3} - \cancel{303n^2} + \cancel{30600n} - 1030000)$$

$$\therefore f(n+1) - f(n) = 3n^2 - 603n + 30298$$

We want to prove

$$f(n) > f(n+1) \longleftrightarrow f(n+1) - f(n) < 0$$

$\therefore$ By solving the quadratic

$$3n^2 - 603n + 30298 < 0$$

$$\therefore b^2 - 4ac = 603^2 - 4(3)(30298)$$

$$= 363609 - 363576$$

$$\boxed{b^2 - 4ac = 33}$$

Roots:

$$n = \frac{603 \pm \sqrt{33}}{6}$$

$$\approx 99.04 \text{ or } 100.96$$

$\therefore$ Thus inequality holds for integers between these roots.

$$\therefore \underline{n = 100}$$

Then, $f(100) > f(101)$

$\therefore$ There exists an integer n such that $f(n) > f(n+1)$.

Q2) Prove that $n^6 = O(n^{\lg n})$ using formal definition of Big-Oh

→ We know that,

$f(n) = O(g(n))$ iff $c > 0$ i.e. constant c is +ve and $f(n) \leq c\,g(n)$ for $\forall n \geq n_0$

Let, $f(n) = n^6$; $g(n) = n^{\lg n}$

Goal :- $\underbrace{(\quad)}_{+ve} n^6 \leq \underbrace{(\quad)}_{+ve} n^{\lg n}$

Comparing the exponents

$\therefore \quad n^6 = n^{\lg 64}$

$\boxed{\because n^{\lg n} = n^{\lg 64}}$

$\therefore$ For all $n \geq 64$

$\lg n \geq \lg 64 = 6$

$\therefore \boxed{n^{\lg n} \geq n^6}$

Let say $c = 1$ $n_0 = 64$ Then for all $n \geq 64$

$n^6 \leq 1 . n^{\lg n}$

$\therefore \boxed{n^6 = O(n^{\lg n})}$ ..... by definition of Big-Oh

Q3) Prove that if $f_1(n) = \Omega(g_1(n))$ and $f_2(n) = \Omega(g_2(n))$ then $f_1(n) + f_2(n) = \Omega(g_1(n) + g_2(n))$ using Big-Ω

→ We know that

$$f(n) = \Omega(g(n))$$

means, there exist constants $c > 0$ and n_0

such that:

$$f(n) \geq c\, g(n) \quad \text{for } \forall n \geq n_0$$

Applying the above definition to each function separately

$$f_1(n) = \Omega(g_1(n))$$

$c_1 > 0$ and n_{01} such that

$$f_1(n) \geq c_1 g_1(n) \quad \forall n \geq n_{01}$$

Similarly,

$$f_2(n) = \Omega(g_2(n))$$

$c_2 > 0$ and n_{02} such that

$$f_2(n) \geq c_2 g_2(n) \quad \forall n \geq n_{02}$$

Let $n_0 = \max(n_{01}, n_{02})$, then for all $n \geq n_0$ both inequalities hold

$$f_1(n) \geq c_1 g_1(n) \quad \& \quad f_2(n) \geq c_2 g_2(n)$$

$\therefore$ Add the inequalities.

for all $n \geq n_0$

$$f_1(n) + f_2(n) \geq c_1 g_1(n) + c_2 g_2(n)$$

Now, for c, Let $c = \min(c_1, c_2)$

$\therefore c \leq c_1$ & $c \leq c_2$, we get

$$c_1 g_1(n) + c_2 g_2(n) \geq c(g_1(n) + g_2(n))$$

Final conclusion

There exist constants $c > 0$ & n_0 such that

$$f_1(n) + f_2(n) \geq c(g_1(n) + g_2(n)) \quad \forall n > n_0$$

Therefore by formal definition of Big-Omega

$$f_1(n) + f_2(n) = \Omega(g_1(n) + g_2(n))$$

Q4) As discussed with the professor this problem is similar to problem 3 and uses the <u>induction hypothesis</u>.

Since, $f_1(n) + \ldots\ldots + f_k(n) = \Omega(g_1(n) + \ldots + g_k(n))$

by definition, there exists constants $c > 0$ & n_0 for $\forall n \geq n_0$

$$f_1(n) + \ldots\ldots + f_k(n) \geq c(g_1(n) + \ldots\ldots + g_k(n)) \quad \text{— (i)}$$

Applying the Big-Omega definit^n to $(k+1)$th term

$$f_{k+1}(n) = \Omega(g_{k+1}(n))$$

where $c_{k+1} > 0$ & n_{k+1} such that for $\forall n \geq n_{k+1}$

$$\cancel{f_{k+1}(n) \geq c_k + g_k + 1(n)}$$

$$\therefore f_{k+1}(n) \geq c_{k+1}(g_{k+1}(n)) \quad \text{— (ii)}$$

Let say $n^* = \max(n_0, n_{k+1})$

then $n \geq n^*$ inequalities hold for both (i) & (ii)

for all $n \geq n^*$

$$f_1(n) + \dots + f_{k+1}(n) = \left[f_1(n) + \dots + f_k(n)\right] + f_{k+1}(n)$$
$$\geq c\left(g_1(n) + \dots + g_k(n)\right) + c_{k+1} g_{k+1}(n)$$

Let $c^* = \min(c, c_{k+1})$

Since $c^* \leq c$ & $c^* \leq c_{k+1}$

$$c\left(g_1(n) + \dots + g_k(n)\right) + c_{k+1} g_{k+1}(n)$$
$$\geq c^*\left(g_1(n) + \dots + g_{k+1}(n)\right)$$

Thus, there exist constants $c^* > 0$ & n^* such that $\forall n \geq n^*$

$$f_1(n) + \dots + f_{k+1}(n) \geq c^*\left(g_1(n) + \dots + g_{k+1}(n)\right)$$

Therefore, by formal definition of Big-Omega.

$$f_1(n) + f_2(n) + \dots + f_{k+1}(n) = \Omega\left(g_1(n) + g_2(n) + \dots + g_{k+1}(n)\right)$$

Q5) Prove, There exists some polynomial function $f(n)$ such that $f(x) > f(x+1)$ & x is the only integer where $f(n) > f(n+1)$.

→ Lets put any integer x.

$$\therefore f(n) = (n-x)^4 + (n-x)^2$$

we know $f(n) \geq 0$ for all intergers $n - f(x) = 0$

as the f^n is symmetric & increasing $|n-x|$

Now, $f(n+1) - f(n)$

The difference is -ve only when $n = x$ (+ve for all integers)

Thus, $f(x) > f(x+1)$

and x is the only integer for which this holds.

Therefore, for every integer x there exists a polynomial funct^n satisfying the required condition.